

DRY-RUN GUIDE | 07_kadanes_algorithm.py

Source: Python\dsa_solutions\07_kadanes_algorithm.py

CORE BEHAVIOR

Tracks the best subarray ending at each position and the best total seen (Kadane's algorithm).

REPRESENTATIVE INPUT

'[-2,1,-3,4,-1,2,1,-5,4]'.

STEP TRACE

The contiguous run 4,-1,2,1 yields running totals 4,3,5,6; global best=6.

EXPECTED OUTPUT / RETURN

'6'

ACTUAL SOURCE EXCERPT

Verbatim excerpt from Python\dsa_solutions\07_kadanes_algorithm.py; line numbers refer to the original file. No source was modified.

```
001 | import sys
002 |
003 |
004 | def main():
005 |     data = list(map(int, sys.stdin.read().split()))
006 |     if not data or data[0] == 0:
007 |         print(0)
008 |         return
009 |     count = data[0]
010 |     numbers = data[1:1 + count]
011 |     current_sum = best_sum = numbers[0]
012 |     for number in numbers[1:]:
013 |         current_sum = max(number, current_sum + number)
014 |         best_sum = max(best_sum, current_sum)
015 |     print(best_sum)
016 |
017 |
018 | if __name__ == "__main__":
019 |     main()
020 |
```

DRY-RUN NOTE

Values are representative. For function-only modules the guide uses an illustrative direct call and explains its return; quirks/errors describe actual behavior.